

Learning Objective 1.1: I can explain what an antenna is, describe its role in a wireless system, and recognize common antenna types by sight.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Name that antenna.** Each description below is what you would see if you walked up to the hardware. For each, name the antenna family and give its *job* — the reason an engineer reaches for that shape — in one or two sentences.
 - (a) **(i)** A straight metal rod, split and fed at its centre, roughly half a wavelength long overall.
(ii) A single rod about a quarter of a wavelength tall, standing on a large sheet of metal or a vehicle roof and fed at its base.
 - (b) A thin rectangle of copper etched on a grounded dielectric board, about half a wavelength across, fed from underneath or from a trace at its edge.
 - (c) A rectangular metal pipe that flares open at one end, fed from a waveguide at the other.
 - (d) **(i)** A row of parallel rods of slightly different lengths mounted along a boom, with only one of them connected to a cable. **(ii)** A large curved metal dish with a small antenna suspended at its focus.

2. **What an antenna is, and what it is for.** Answer each part in one or two complete sentences.

- (a) Define an antenna as a *transducer*. Name the two kinds of wave it sits between, and say which component in an RF link block diagram (source → amplifier → transmission line → antenna → channel) is the only one that touches free space.

- (b) Antennas do not create energy. Given that, explain what a vendor means by “this antenna has 30 dBi of gain.”

- (c) A tracking antenna is measured on transmit and found to have a 3° beam pointed at boresight. You now use the same antenna to *receive*. State what reciprocity tells you about the receive pattern, and give two practical consequences for the receiving station.

- (d) Match each band to the mission it is usually flown for, and state in one sentence what sets the size of the antenna in each case. **Bands:** (i) 3–30MHz (HF), (ii) 225–400MHz (UHF), (iii) 1575MHz (GPS L1), (iv) 8–12GHz (X-band). **Missions:** fire-control and imaging radar; beyond-line-of-sight skywave communications; navigation; tactical manpack radio and UHF SATCOM.

Documentation: _____